



STUDENT PROJECTS

PROJECT PLAN

Department:	EC	Semester:	6 th	Academic Year:	2026– 27
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A. Proposed Title of the Project:

Anviksha 8-Bit CPU

B. Name(s) of guide(s): Prof. / Dr. / Mr. / Ms.

C. Name of Team Members (Strictly not more than four students in a batch):

USN	Name	
4VP23EC050	Mr. THIYANAND M	Team Leader
4VP23EC005	Ms. CHAITHANYA	Team Member
4VP23EC010	Ms. FATHIMATH ANEESHA	Team Member
4VP23EC029	Ms. PREETHA S M	Team Member

D. Date of commencement of project:

E. Probable date of completion of the project:

F. If this project is an **improvement** / **extension** of an earlier project, please give the title of earlier project and the year in which it was carried out. Specify the improvements / modifications clearly:

Not Applicable

G. Objectives of the project highlighting its importance (not more than half a page):

- To design and implement a functional 8-bit CPU using basic digital electronics components.
- To understand the instruction, fetch, decode, and execute cycle in a processor.
- To integrate core CPU blocks such as ALU, registers, program counter, control unit, and memory.
- To analyze data flow, control signals, and clock timing within the CPU.
- To bridge theoretical computer architecture concepts with practical hardware implementation

H. Block diagram and Methodology (Not more than a page):

The device operates in the following sequence:

- Power Supply Setup**
A regulated 5V power supply is designed and tested to ensure stable operation of all ICs used in the CPU.
- Clock Generation**
A clock circuit is implemented to provide synchronized timing pulses required for instruction execution and data transfer.
- Data Bus Design**
An 8-bit data bus is created to allow controlled communication between registers, ALU, memory, and output units.
- Register Implementation**
Registers are designed to temporarily store data and operands needed during processing.
- ALU Design**
The Arithmetic Logic Unit is developed to perform basic arithmetic and logical operations on 8-bit data.

Prepared by:

Checked by:

HOD

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6. Instruction Control

The Program Counter and Instruction Register are implemented to manage instruction sequencing and hold current instructions.

7. Control Unit Design

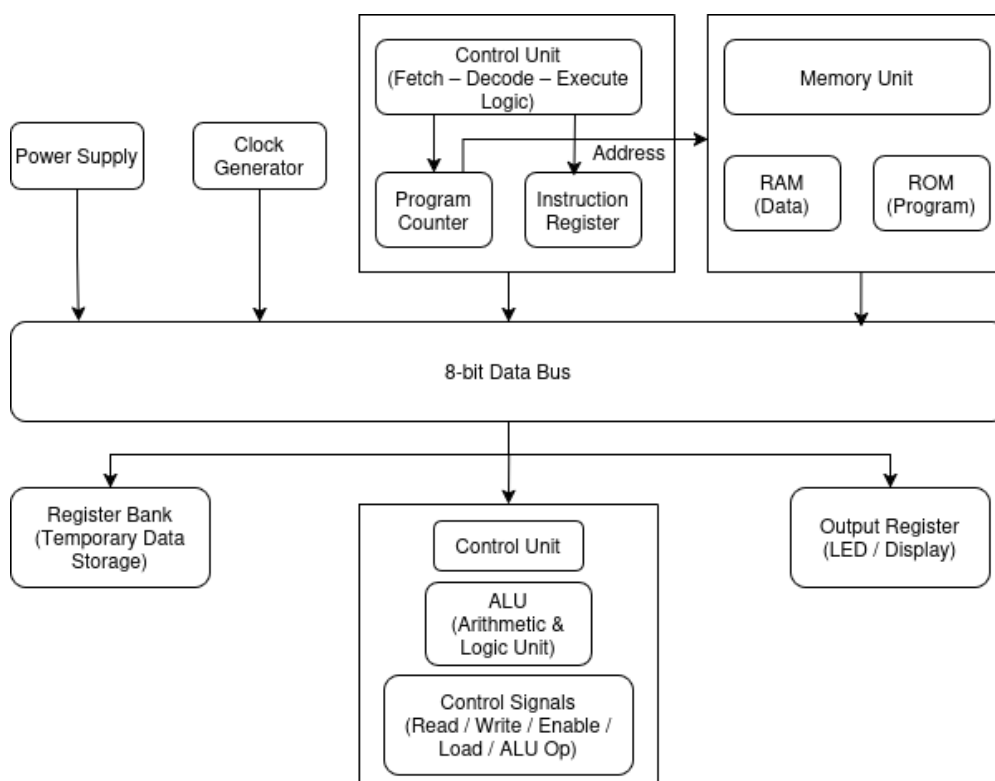
A control unit is designed to generate required control signals for fetch, decode, and execute cycles.

8. Memory Integration

ROM and RAM are interfaced to store programs and data required for CPU operation.

9. Output and Testing

An output register is added to display results, and the complete CPU is tested for correct functionality.



I. Expected Outcome of the project

- Successful design and implementation of a fully functional 8-bit Central Processing Unit.
- Correct execution of instruction fetches, decode, and execute cycles.
- Reliable performance of core CPU components such as ALU, registers, program counter, control unit, and memory.
- Accurate execution of basic arithmetic, logical, and data transfer instructions.
- Proper synchronization of operations using the clock signal.
- Clear visualization of outputs through the output register and LEDs.
- Improved practical understanding of computer architecture and digital processor design.



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J. Application of the project:

- Used as an educational model to understand basic CPU architecture and internal working.
- Helps in learning instruction execution, data flow, and control logic of processors.
- Can be applied in simple embedded systems with low computational requirements.
- Serves as a foundation for designing microcontrollers and advanced CPUs.
- Useful for digital electronics and computer organization laboratory demonstrations.

K. Does the project proposed is relevant to any of the Industry or Institution in and around your area (Yes / No):

No

L. If yes, please provide details of the industry / institution and contact details:

M. Budget:

Materials Cost:8500/-

Labour Charges:

Any other cost:

Source of Funds:

Self / Industry / Others

N. Expected Completion Time:

Expected Time for Major Activities

Schedule

Synopsis (Project Plan) submission to the Department

Presentation of synopsis

Last date for re-submission (if applicable) & review

Submission of Reports

Internal Viva and/or Project Exhibition

O. Authors:

Student(s) / Author(s)

Signature with date

1.Thiyanand M

2.Chaithanya

3.Fathimath Aneesha

4.Chaithanya

Checked By:

Signature with date:

P. Guidance:

Guide (s) allotted:

Signature (s) with date

1. Guide: Prof. / Dr. / Mr. / Ms.

2. Co-Guide (if any): Prof. / Dr. / Mr. / Ms.